Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Gate interface

File: gate_if.kicad_sch

Current sense, overcurrent, temperature

File: analog.kicad_sch

Resolver / encoder

File: resolver.kicad_sch

Vehicle connector and I/O

File: vehicle.kicad_sch

WiFi (ESP32)

File: wifi.kicad_sch

Replacement logic board for the Chevrolet Bolt EV inverter (OpenInverter stm32-sine, HW_BOLT)

Bolt_Invt open-source inverter controller

Sheet: /

File: bolt_logic.kicad_sch

Title: Bolt logic board

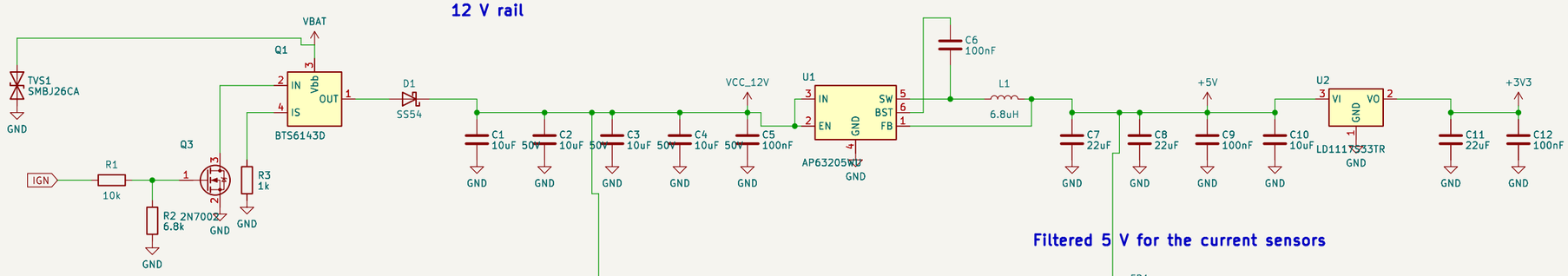
Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/8

Battery in: TVS first, ignition high-side switch Q1, reverse-polarity diode D1

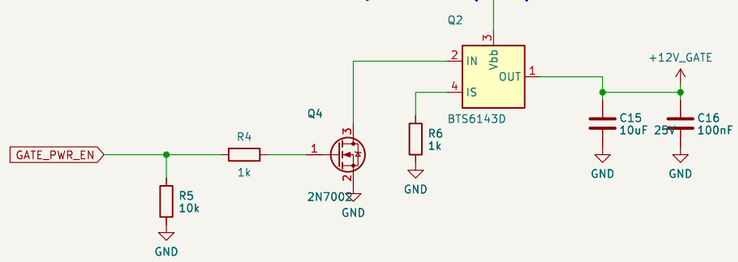
12 V rail

5 V buck (AP63205, 3.8–32 V in)

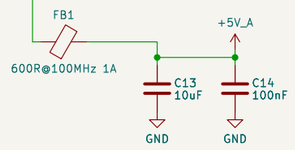
3.3 V LDO (MCU, analog)



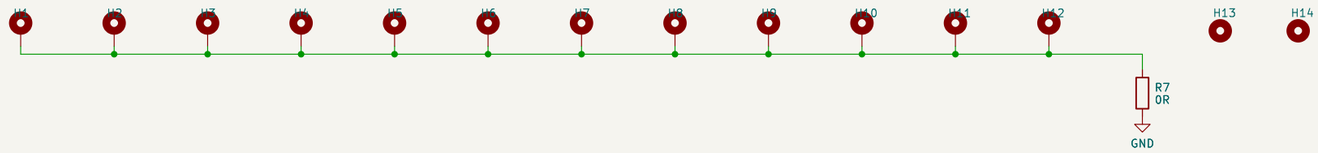
Gate-board 12 V, switched by the MCU (PA15)



Filtered 5 V for the current sensors



Mounting holes: chassis, tied to GND through R7





GENERATED by tools/gen_sch.py from tools/sch_design.py + tools/sch_layout.py — edit those, not this

Bolt_Invt open-source inverter controller

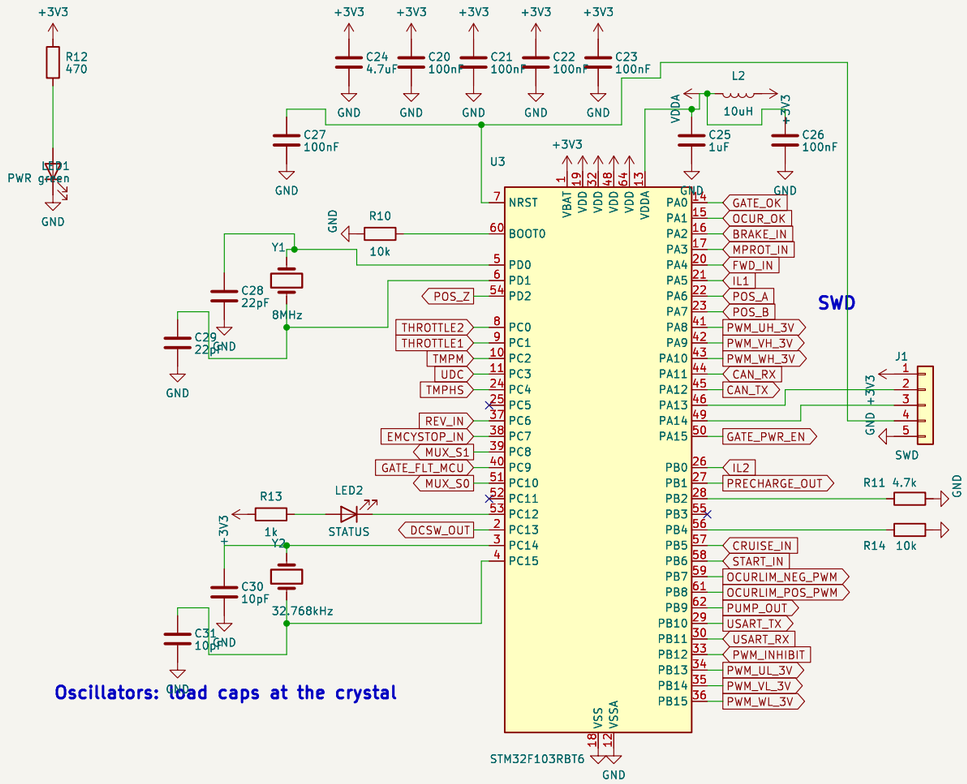
Sheet: /Power/
File: power.kicad_sch

Title: Bolt logic board — Power

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/8

STM32F103RBT6 (OpenInverter stm32-sine, HW_BOLT)

Decoupling: 100 nF at each VDD pin, 4.7 uF bulk



Oscillators: load caps at the crystal

SWD

GENERATED by tools/gen_sch.py from tools/sch_design.py + tools/sch_layout.py — edit those, not this

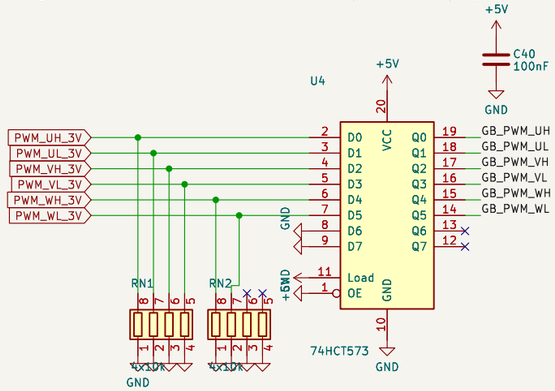
Bolt_Invt open-source inverter controller

Sheet: /MCU/
File: mcu.kicad_sch

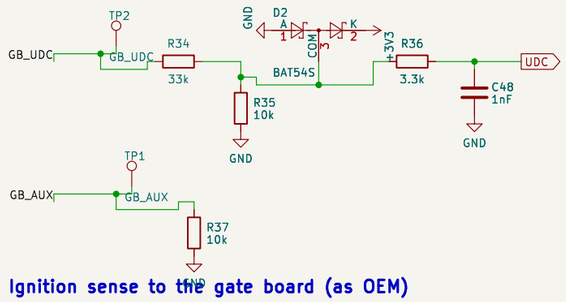
Title: Bolt logic board — MCU

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/8

PWM to the gate board: 3.3 V -> 5 V latch (74HCT573), pulled low while the MCU is in reset

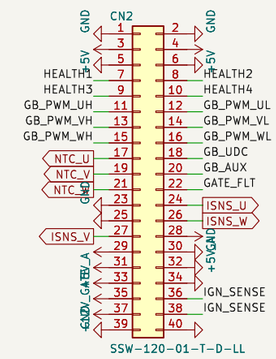


DC-bus voltage from the gate board (CN2 18), clamped to the ADC range

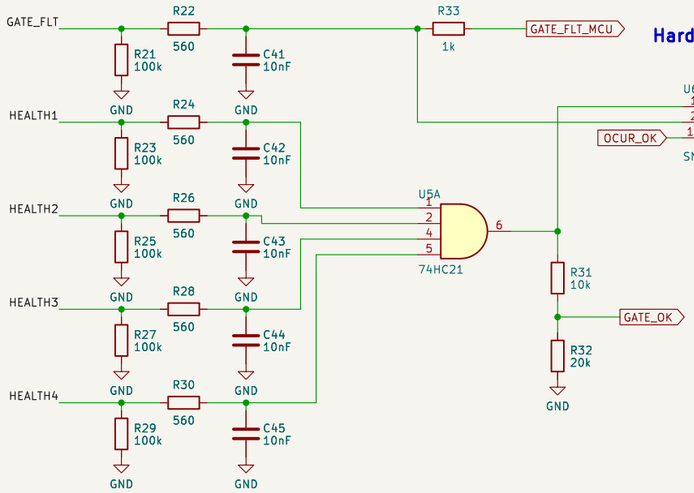


Ignition sense to the gate board (as OEM)

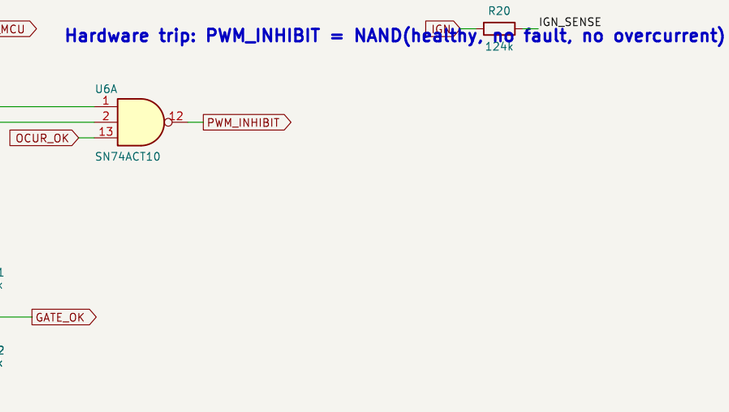
CN2: gate-board connector (2x20)

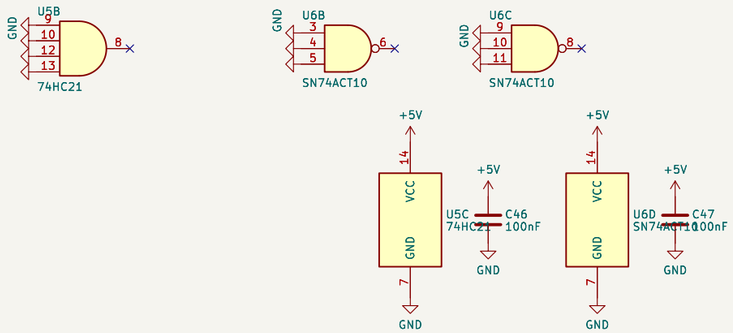


Gate-board status: 100k fail-safe pull-down, 560R / 10 nF filter



Hardware trip: PWM_INHIBIT = NAND(healthy, no fault, no overcurrent)





GENERATED by tools/gen_sch.py from tools/sch_design.py + tools/sch_layout.py — edit those, not this

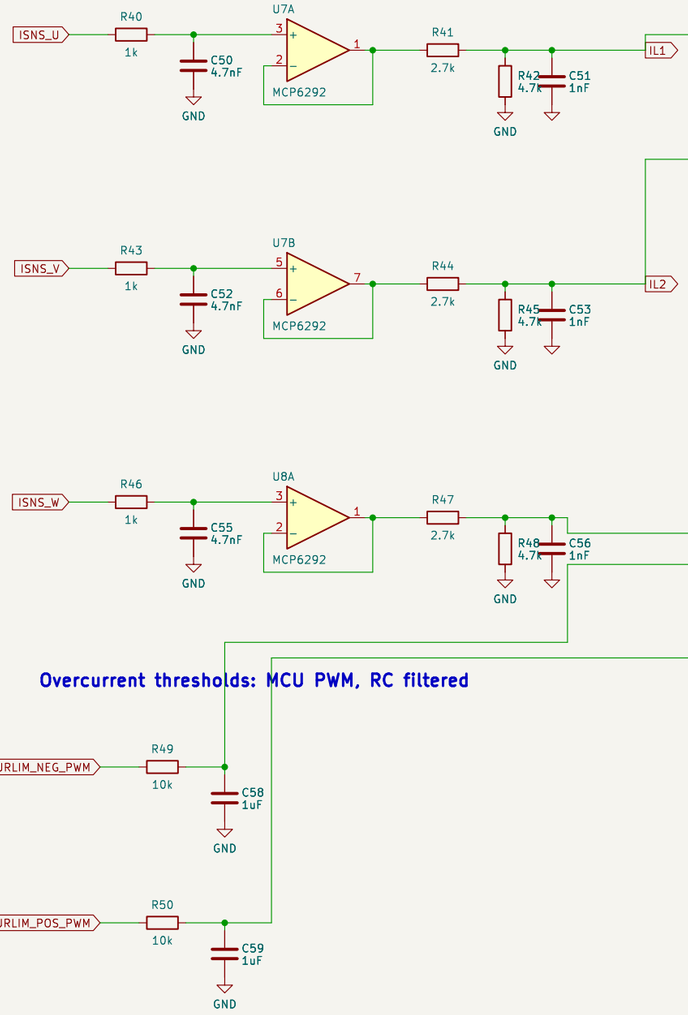
Bolt_Invt open-source inverter controller

Sheet: /Gate interface/
File: gate_if.kicad_sch

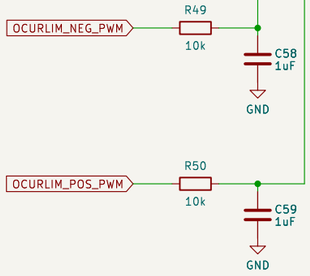
Title: Bolt logic board — Gate interface

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/8

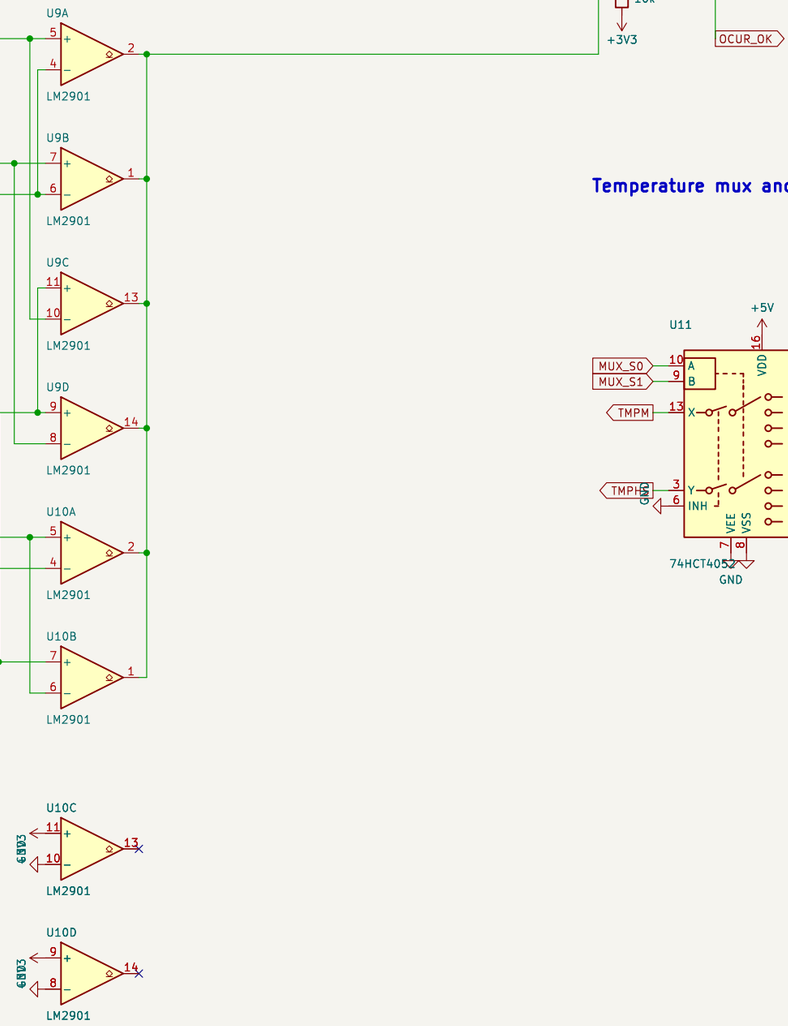
Phase current: RC filter, buffer (5 V_A), 2.7k / 4.7k to the ADC



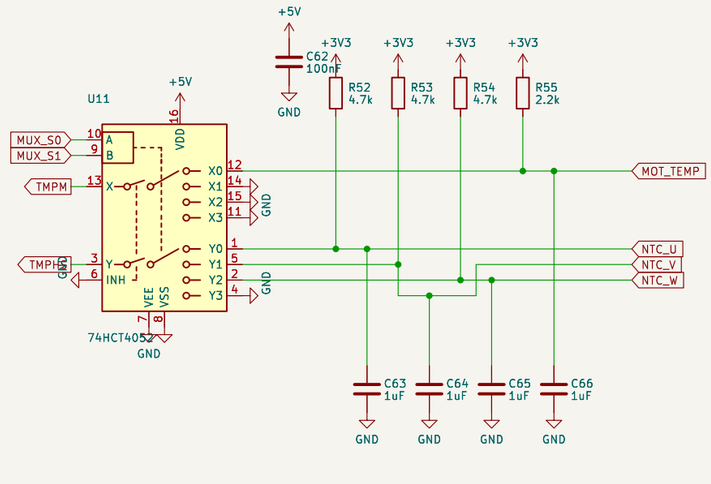
Overcurrent thresholds: MCU PWM, RC filtered

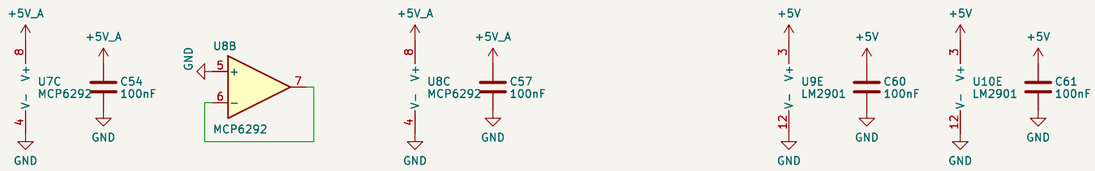


Overcurrent window (LM2901), wired-OR



Temperature mux and NTC pull-ups





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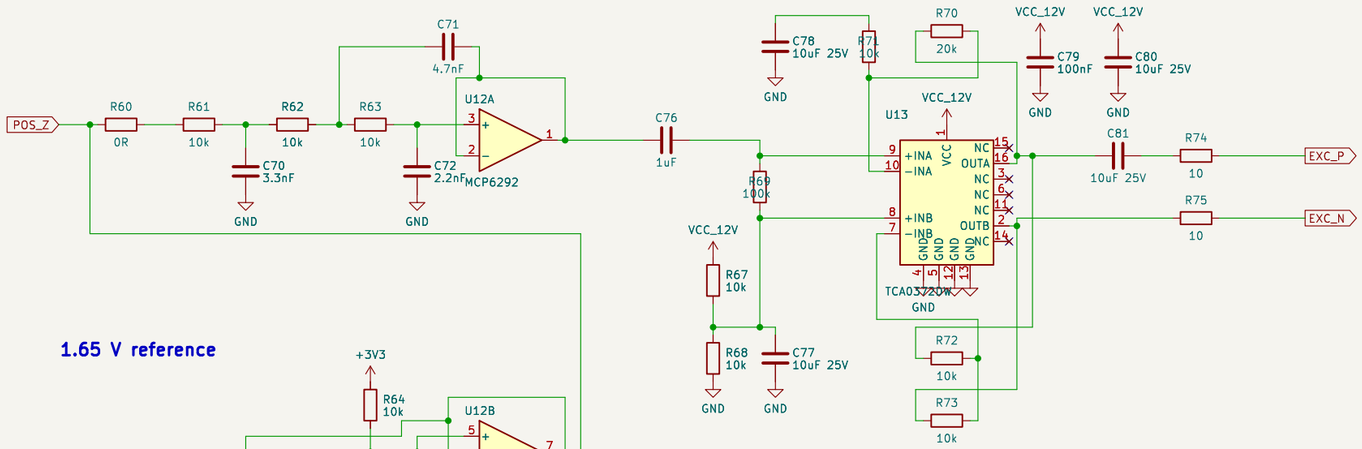
Bolt_Invt open-source inverter controller

Sheet: /Current sense, overcurrent, temperature/
File: analog.kicad_sch

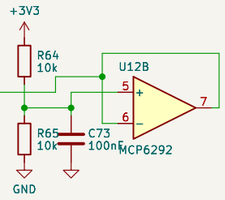
Title: Bolt logic board — Current sense, overcurrent, temperature

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/8

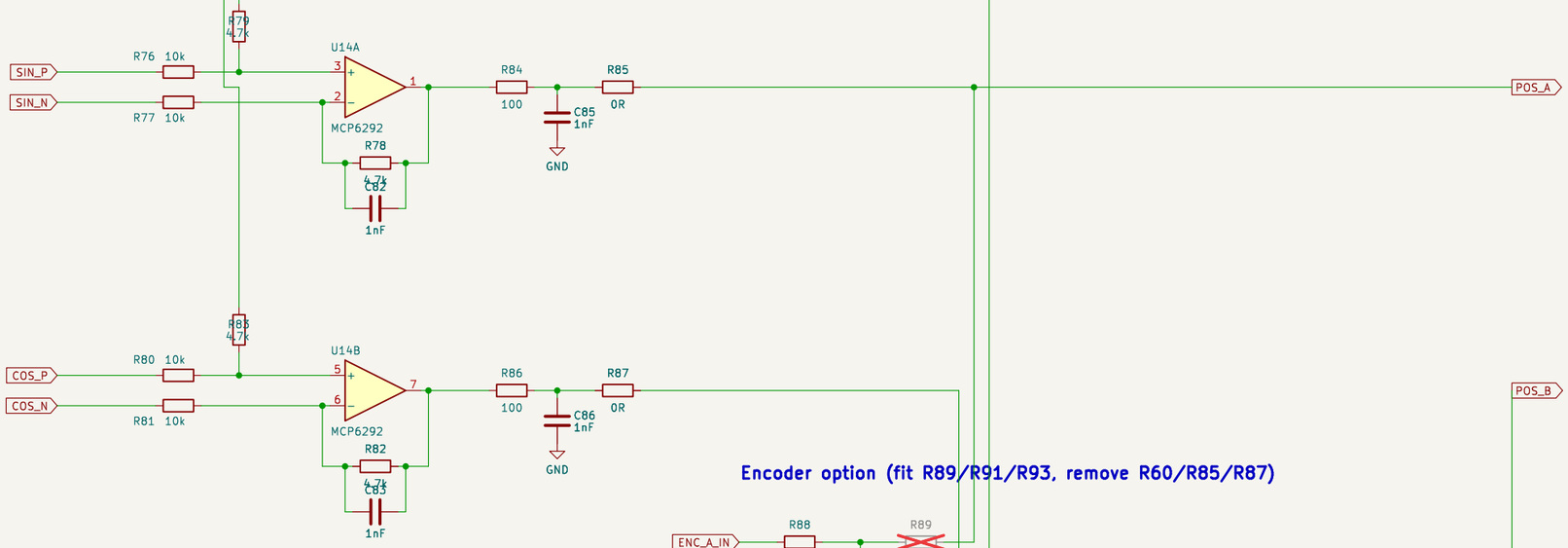
Resolver exciter: MCU square wave -> filter -> sine -> TCA0372 bridge (12 V)



1.65 V reference

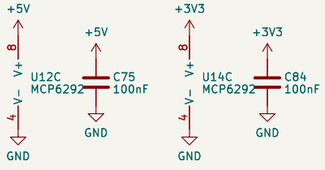


Sin / cos receivers (differential, 3.3 V)



Encoder option (fit R89/R91/R93, remove R60/R85/R87)





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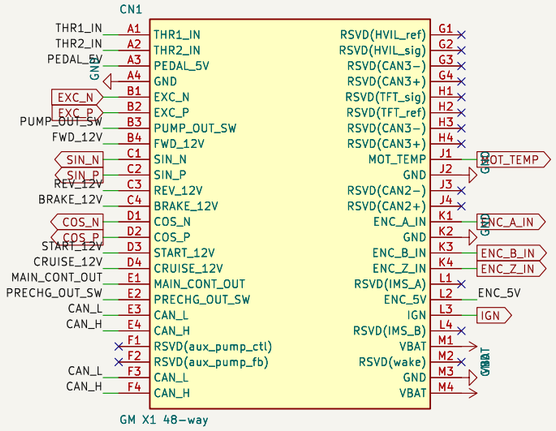
Bolt_Invt open-source inverter controller

Sheet: /Resolver / encoder/
File: resolver.kicad_sch

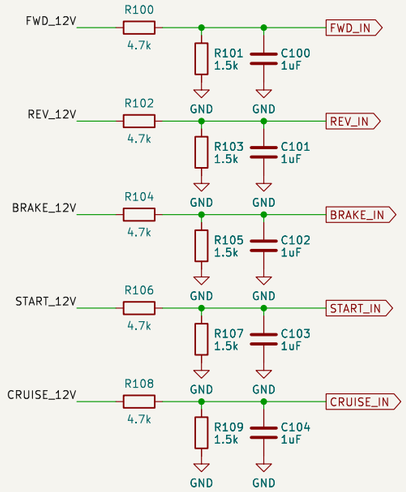
Title: Bolt logic board — Resolver / encoder

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/8

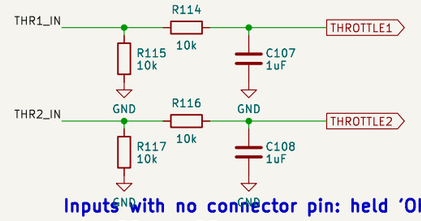
CN1: vehicle connector (GM X1, 48-way)



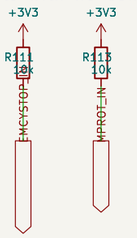
12 V switch inputs: 4.7k / 1.5k divider, 1 uF filter



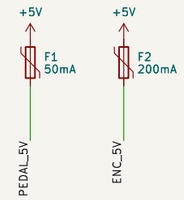
Throttle inputs (0-5 V -> 2.5 V max)



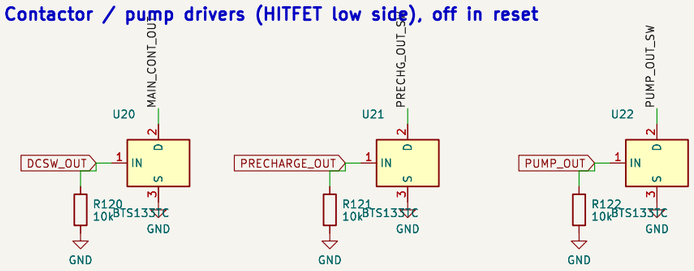
Inputs with no connector pin: held 'OK'



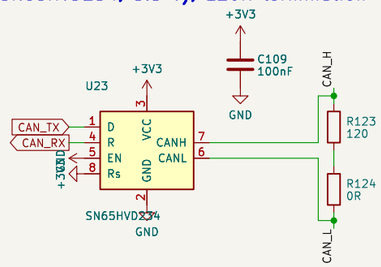
Sensor supplies (polyfuses)



Contactor / pump drivers (HITFET low side), off in reset



CAN (SN65HVD234, 3.3 V), 120R termination



GENERATED by tools/gen_sch.py from tools/sch_design.py + tools/sch_layout.py — edit those, not this

Bolt_Invt open-source inverter controller

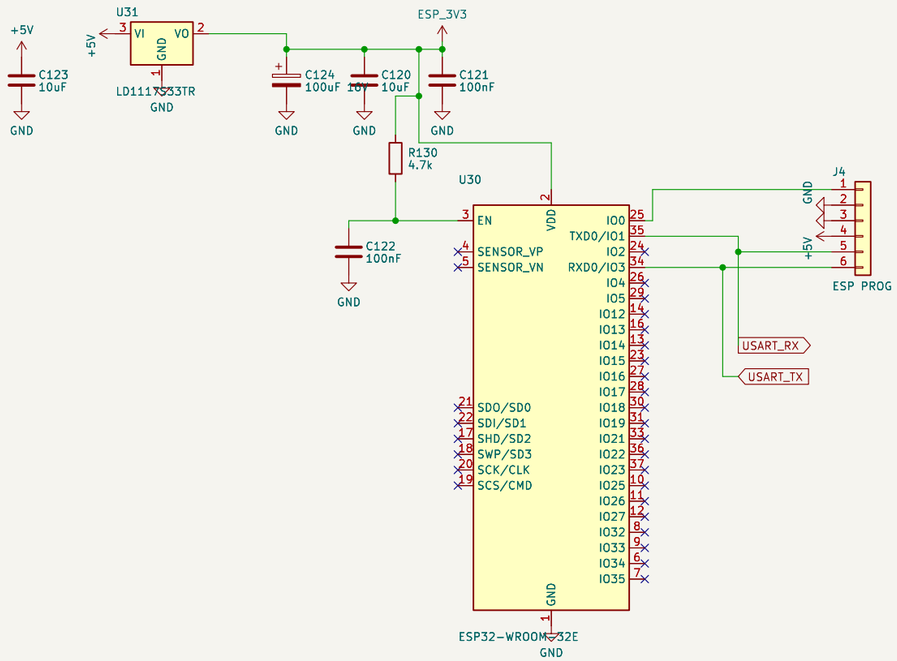
Sheet: /Vehicle connector and I/O/
File: vehicle.kicad_sch

Title: Bolt logic board — Vehicle connector and I/O

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/8

ESP32-WROOM-32E (PCB antenna, as Tesla V6)

Own 3.3 V LDO (forum review: ~0.5 A transmit bursts), 100 uF bulk + 10 uF / 100 nF at the module pin



GENERATED by tools/gen_sch.py from tools/sch_design.py + tools/sch_layout.py — edit those, not this

Bolt_Invt open-source inverter controller

Sheet: /WiFi (ESP32)/
File: wifi.kicad_sch

Title: Bolt logic board — WiFi (ESP32)

Size: A3	Date: 2026-11-06	**Rev: 0.2a**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/8